

Owen Lee

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EDUCATION

The University of Illinois Urbana-Champaign

August 2026 – *Expected* May 2028

Master of Computer Science

The Chinese University of Hong Kong, Shenzhen

September 2022 – July 2026

Bachelor of Engineering, Computer Science and Engineering

GPA: 3.2/4.0

SKILLS

Languages: Python, TypeScript, SQL, Java, C++, JavaScript

AI & ML: LLMs, Retrieval Augmented Generation, RAG, Finetuning with LoRA and QLoRA, NLP, Generative AI

ML Frameworks: PyTorch, HuggingFace Transformers, PEFT, NumPy, scikit-learn, Weaviate, ChromaDB

Technologies & Infra: Spring Boot, REST APIs, AWS EC2, AWS Lambda, AWS Bedrock, Docker, Git, Linux

Data & Visualization: Plotly, Plotly Dash, D3.js, pandas, Matplotlib

WORK AND RESEARCH EXPERIENCE

Capstone Student Researcher

August 2026 – Present

CCC Intelligent Solutions

Chicago, IL

- Traced agent harness execution failures to mine recurring weakness patterns, proposed targeted fixes, and landed them as PRs under research team review

Research Assistant under Prof. Xiaoying Tang

November 2025 – May 2026

The Chinese University of Hong Kong, Shenzhen

- Engineered 1,040 step grounded visual and text assembly traces for the SoT-26K training corpus, supplying supervision for a visual chain of thought VLM accepted to ICML 2026
- Lifted component numeracy to 88.4%, up 24.2 pts, and structural topology to 84.8%, up 19.3 pts, over direct generation via chain of thought templates aligning rationales to rendered intermediate states

Artificial Intelligence Engineering Intern

July 2025 – October 2025

ATG Nexus Sdn Bhd

Puchong, Malaysia

- Shipped a hybrid keyword and semantic retrieval engine on Weaviate and OpenAI embeddings that replaced keyword only search in a client facing legal document search product
- Increased analyst throughput 50% by automating writeups with a bilingual Gemini report generator and Plotly pipeline

Research Assistant under Prof. Benyou Wang

January 2025 – May 2025

The Chinese University of Hong Kong, Shenzhen

- Reduced caption hallucination 15% by building a multimodal crawler and CLAP filtered curation pipeline that assembled 1.2M audio caption pairs
- Beat all 5 baselines at 44.3 text to audio and 57.8 audio to text R@1 with an HTSAT+BERT retrieval model trained on an EC2 scaled S3 pipeline, accepted to ACL 2026

Software Engineering Intern

May 2024 – August 2024

China Unicom

Zhejiang, China

- Developed a Java Spring Boot REST API serving real time weather dashboards at 95% success across about 2,400 requests per hour, plus a health monitor hitting 99% alert detection accuracy

PROJECT EXPERIENCE

MoE Routing vs. Dense Finetuning vs. Zero Shot API | *PyTorch, HuggingFace, QLoRA*

Aug 2026

- Benchmarked a 2B QLoRA specialist router against a dense QLoRA 4B model on 1,350 graded examples, isolating reasoning depth, as the bottleneck, with dense leading by about 30% even at 100% router precision

Redline: LLM Red Teaming Harness | *FastAPI, Next.js, AWS Bedrock*

Aug 2026 – Sep 2026

- Built a red teaming harness probing unsafe behavior across 4 AWS Bedrock models and 8 attack types, including prompt injection, jailbreaks, and reasoning traps
- Exposed a refusal training gap by automatically scoring 800 responses with an LLM judge pipeline: models blocked 88 to 100% of direct harmful requests but only 24 to 84% of indirect injections

Retrieval Grounded Text to SQL Chatbot | *Python, MySQL, ChromaDB, SQLGlott*

October 2025

- Architected a schema aware RAG pipeline on all-MiniLM and ChromaDB grounding SQL generation in live database metadata, eliminating handwritten schema prompts
- Measured 89% execution accuracy with an evaluation harness running 500+ generated queries against a live MySQL instance and diffing result sets

PUBLICATIONS

Y. Huo, S. Zhang, K. Zeng, H. Liu, **O. Lee**, J. Chen, Y. Lu, Y. Guo, Y. Liang, X. Tang. *Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought*. Paper. International Conference on Machine Learning, ICML 2026.

S. Chen, X. Yuan, Z. Li, L. Zhao, **O. Lee**, Z. Su, Q. Sun, B. Wang. *FusionAudio-1.2M: Towards Fine-grained Audio Captioning with Multimodal Contextual Fusion*. Paper. Association for Computational Linguistics, ACL 2026.